



Harnessing water hyacinth for biogas and bioslurry generation: experimental insights on biomass characterization and crop yield enhancement

Tewodros Mulu^{a,*}, Bilhate Chala^b, Yezbie Kassa^a, Tayachew Nega^c, Marelgiln Adugna^d, Mohamed Gedefaw^d, Teferi Alem^d, Kiflie Tesfamariam^e, Tadesse Adgo^{f,h,i}, Beatriz Waldman^g, Bernhard Freyer^{h,i}, Dessie Tibebe^{a,*}

^a College of Natural and Computational Sciences, University of Gondar, P.O. Box 196, Gondar, Ethiopia

^b University of Hohenheim | Institute of agricultural engineering (440e), Germany

^c Institutes of Technology, University of Gondar, and P.O.BOX 196, Ethiopia

^d College of Agriculture and Environmental Sciences, University of Gondar, P. O. box 196

^e Tana and other water bodies Authority, Maksegnit, Ethiopia

^f The Nature and Biodiversity Conservation Union Ethiopia (NABU Ethiopia) | Bahir Dar city, Tana Sub city, Shimbit Kebele, P.O. Box 503, Ethiopia

^g The Nature and Biodiversity Conservation Union (NABU e.V.) | Charitéstraße 3 10117, Germany

^h College of Agriculture and Environmental Sciences, Bahir Dar University, P. O. box 176, Ethiopia

ⁱ University for continuing education Krems, Faculty of Business and Globalization, Kreis, Austria

ARTICLE INFO

Editor: DR B Gyampoh

Keywords:

Biogas

Bio-slurry

Water hyacinth

Batch

Semi-continuous

Anaerobic digestion

ABSTRACT

This study explored the physicochemical characterization of several components of water hyacinth and its biogas potential in batch and semi-continuous mode. The effect of the bioslurry from the continuous digesters were evaluated on the crop biomass yield. In the batch studies, several portions of water hyacinth, such as complete fresh biomass leaf stem and root, were evaluated to identify the most effective components for biogas and bioslurry generation.

The semi-continuous trials used two stainless steel digesters, each with a capacity of 650 L, to produce biogas and bioslurry. The anaerobic digestion process required semi-continuous feeding of substrate and other materials, as well as regular monitoring of gas flow rate, temperature, total solid, volatile solid, and pH levels. The generated biogas and bioslurry were thoroughly characterised.

The results reveal that the whole fresh biomass has substantial potential for biogas and bioslurry generation. The fresh stem produced the highest biogas and methane yields, with values of 597 and 326 LN/kgVS (litres normalised per kilogramme of volatile solids), respectively. The biogas exhibited an average methane and carbon dioxide quality of $59.2 \pm 0.2\%$ and $38.9 \pm 0.2\%$, respectively, with a minimum concentration of 1.0 ± 0.8 ppm for hydrogen sulphide. Bioslurry application yields the highest yield ($169.43 \text{ kg/ha} \pm 3.5$) for Teff crops, followed by mineral fertiliser ($159.43 \text{ kg/ha} \pm 5.03$) from a 0.0006 hectare area. As a result, our findings provide useful insights into the practical application of water hyacinth as a sustainable source of biogas generation and bioslurry as an organic fertiliser to increase agricultural productivity.

* Corresponding authors.

E-mail addresses: Tewodros.mulu@uog.edu.et (T. Mulu), dessie.tibebe@uog.edu.et (D. Tibebe).

<https://doi.org/10.1016/j.sciaf.2025.e02730>

Received 17 September 2024; Received in revised form 18 April 2025; Accepted 2 May 2025

Available online 8 May 2025

2468-2276/© 2025 The Author(s). Published by Elsevier B.V. This is an open access article under the CC BY-NC-ND license (<http://creativecommons.org/licenses/by-nc-nd/4.0/>).

Introduction

Energy supply is significant for the economic development of a country; energy plays an important role, especially now all human activities are very dependent on it. Energy is used for supporting various devices, including lighting equipment, household appliances, and industrial machinery. Hence, result from technological advances and global warming, successively promoting the utilization of renewable energies in circular economy approach [1].

The need for alternative sources of energy generation that are inexpensive, ecofriendly, renewable and can replace fossil fuels is on, owing to the increasing demands of energy. One approach in this direction is the conversion of plant residues and weeds into biofuels wherein lignocellulose, which forms the structural framework of plants consisting of cellulose, hemicellulose and lignin, is first broken down and hydrolyzed which upon anerobic digestion form biogas [2].

Water hyacinth (*Eichhornia crassipes*) is a weed ubiquitously found in lakes, rivers, and other water bodies across the globe [3]. It is traditionally considered as a weed and is found to affect water bodies across the globe. The extremely rapid growth and ability to persist in water has made water hyacinth a notorious crop that is difficult to control [4]. It causes a series of damages, including blockage of irrigation channels affecting the flow of water to fields, hindrance of water transport and fishing, affecting the balance of the local ecosystem. All these have a large impact on the livelihoods of marginal farmers, and can lead to poverty (Paul Njogu1, Robert Kinyua1, Purity Muthoni [8]).

Lake Tana is the largest freshwater body in Ethiopia. It constitutes 50 of countries fresh water and has substantial socio-ecological importance. The lake provides important ecosystem services and contributes significantly to food security, livelihoods, and national economies through direct exploitation of fisheries, water resources for irrigation, the recharging of groundwater, hydropower generation, recreation, transport, tourism, microclimate stabilization, climate regulation, and as a home for birds and wildlife. Several monasteries located on the lakes island hold important historical and religious assests.in recognition of the lake's rich biodiversity, UNESCO has recognised the lake in terms of its World Network of Biosphere Reserves in June 2015 [5]. Lake Tana is also the source of the Blue Nile.Despite its ecological and economic importance, both locally and globally, Lake Tana has been facing alarming environmental degradation and loss of biodiversity due to the growth of aquatic weeds such as water hyacinth (WH). The presence of invasive WH in Lake Tana was recognized in 2011 and, in 2014, about 20,000 ha of the north-eastern shore of the lake Tana was infested. It currently infested about 50,000 ha. There is increasing concern about the consequences of the weed encroachment, which has been recognised as an ecological catastrophe for the lake ecosystems and for its ability to supply ecosystem services to the communities whose livelihoods depend on the existence of the lake. Faced with such major changes, there is an urgent need for knowledge of mapping and monitoring how the ecosystems services of the lake are affected by this ecological catastrophe.

Methods to control WH have been proposed, such as biological, and chemical eradication, and physical removal by manpower or machine. Each method has its own benefits and drawbacks [5]. Given the economic and environmental problems involved, the Ethiopian government has attempted to control the weed by mobilising affected farming communities to remove it using manual labour. Currently, a mechanical harvester machine is used to chop, harvest, and dispose of the weed. The WH biomass is usually burned or thrown onto the banks in piles, which results in it rotting. When this happens, methane gas, which contributes to climate change, is released. These fragmented efforts are inefficient in controlling the spread of the weed and its economic, social, and ecological consequences. Thus far, there is no attempt to transform the WH biomass into value-generating products such as eco-friendly handicrafts, energy, or agricultural inputs. Assess the potential intervention options for transforming the WH biomass into value generating

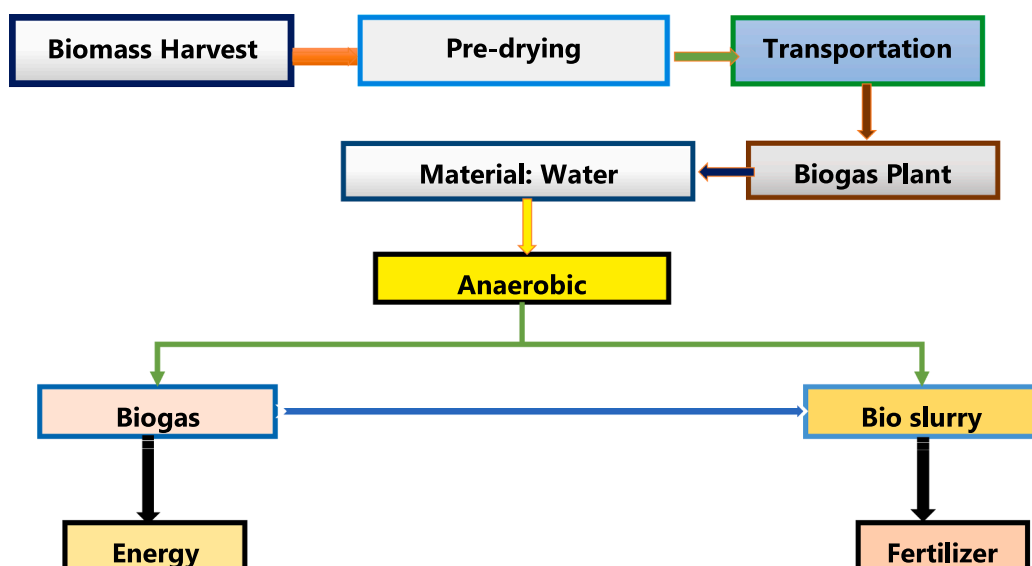


Fig. 1. Overview of the utilization of WH biomass.

products (biogas & bioslurry).

One approach to solve this problem involves the use of Anaerobic digestion (AD). It is described as a biological degradation of organic materials without the presence of oxygen, thereby resulting in methane-rich biogas and an enriched digestate. The methane generated during AD can be converted to different energies depending on indigenous needs; it can be compressed and liquefied as a transportation fuel, generation of electricity, heat and cooking. Whereas, the digestate improves the soil fertility and reduces the load and cost of chemical fertilisers. AD also helps for sustainable waste management opportunity by reducing organic wastes on landfills and economically feasible wastewater treatment [1]. Seven out of the eight variables that form the environmental pillar, namely GHG emissions, soil quality, non-GHG air pollutants, water use and efficiency, water quality, biological variety in the environment and land use can be improved by biogas production. Given the enormous benefits of waste management, Biogas has been the focus of renewed attention due to environmental credits and co-product value gain. Government and non-government incentives, as well as laws, policies, and regulatory frameworks, have played a significant role in this revival [6].

Hence, developing a strategy for sustainable utilization of this weed is considered to be very relevant to local communities [7]. Nevertheless, the intervention also supports farmers for generating extra income by producing more crop yield and sustainable energy supply in case of biogas plant [8]. According to Fadaïro et al. [9], biogas can be produced from WH biomass, especially when blended with animal wastes. Due to its high water content and an appropriate C/N ratio, WH can be used to produce biogas through anaerobic digestion [10]. It can be degraded easily and gives a high gas yield, which is around 340 L/kg DM, much higher than that from rice straw and other crop residuals. Bioslurry is another important by-product, which can seriously increase crop yield through its fast nitrogen effect, on the other hand the crop residues contribute to soil carbon.

The bioslurry from anaerobic digestion can be utilised as organic fertiliser. Because of its abundance and high C/N ratio, WH can be exploited as a possible fuel for biogas generation (Fig. 1) [11].

Several studies show that the sole application of both bio-slurry and chemical fertilizers increase the agronomic performances of crops when compared to no fertilizer application [12]. Application of bio-slurry provides both macro and micronutrients and increases growth and yield variables of field crops, including Teff.

Teff (*Eragrostis tef* (Zucc.) Trotter) is one of the most important cereal crops, and Ethiopia is regarded as the origin and diversity of the crop [13]. Teff is well adapted to a variety of agro-ecological zones, including marginal areas where most crops cannot be grown [14]. Despite its small seed size, farmers prefer to grow Teff due to its numerous advantages, including high market value, low postharvest management costs and low risk. The straw is also preferable animal feed as compared to other cereal crops.

Therefore, the objective of this study was: i. to characterize the chemical composition of water hyacinth and the produced biogas, ii. to evaluate the efficiency of the produced biogas, iii. to determine the effect of organic loading rates for biogas production iv. To compare the effect of WH bio-slurry with inorganic fertilizers on soil amendment, growth and yield on teff.

Methodology

Physicochemical and nutrient analyses of water hyacinth

All the physicochemical characteristics of substrates were evaluated using the method described by the Association of Official Analytical Chemistry (AOAC). Total solids (TS) were measured using weight loss and continued up to evaporation at 105 °C throughout a full day. The weight loss after ignition at 550 °C for 2 hours in a muffle furnace was used to compute the volatile solid (VS) content. The ash content, 1 g of dried Soke sample, was put in a crucible and placed in a muffle furnace (BIBBY, R0000, UK) at 550 °C for 3 h; then it was cooled in a desiccator and weighed. The pH of the feedstock and its bio-slurry was measured using digital pH meter (HI160, Hach, Germany) according to FAO [15], 2021c.

Laboratory analyses were conducted by following standard methods. pH and EC were measured. Organic carbon (OC) and total nitrogen (TN) content were determined by measuring colorimetrically following the standard procedures [15] and Kjeldahl digestion method according to Sertsu & Bekele protocols, respectively. Available phosphorus was extracted by 0.5 sodium bicarbonate solution as described by Olsen method [16], and thereafter measured using spectrophotometer at 880 nm. Total Phosphorus (TP) in bioslurry sample was analyzed by following protocol. Sulfur was determined in the WH plant sample extract by the monocalcium phosphate extracting method using spectrophotometer at 440 nm according to [17].

Total metal analyses in plant sample

The air-dried plant samples were grinded with electrical grinder. Transfer 1 g powdered sample in 100 ml conical flask for acid digestion. Digestion of plant samples was performed based on [18] standards. A total of 13 mL that of 10 ml conc. HNO₃ (70 %), and 3 ml conc. HClO₄ (70 %) were mixed into a 100 ml conical flask with 1 g of plant samples. The samples were digested on hot plate at 120 °C for 2 hr. The clear solution was filtered with 50 mL volumetric flask and filled with deionized water up to the mark. The filtrate stored at 4 °C for heavy metal analysis using FAAS.

Digester design and development

Due to the fact that there is not any equipment for the research available in the study area, a pilot scale digester was developed, equipped with all relevant instruments such as manual stirrer, heat mats, temperature controller, the biogas volume was measured every day at 9:00 AM by using gas flow meter (Gas Meter-JBD 2.5-SA, China), biogas composition was measured every with in two days in triplicate using gas analyser (Biogas 5000; Geotech, Germany) [19] and gas storage bags (1000 L). Two pilot scale digesters were

built with vertical cylinder shape from stainless steel material. The internal dimensions were 910 mm of diameter and 1000 mm of height. The active volume of the digesters was 600 L (Fig. 2). The digesters are equipped with mechanical stirrer and heat mats (240 W, each) fixed from outer surface of the digesters to maintain the fermentation temperature at fixed value which is controlled by a thermostat and temperature sensor (H-Tronic TS 125). The biogas was measured continuously with a gas flow meters connected with the digesters. The biogas afterwards is collected in a gas bag with capacity of 1000 L. The ambient temperature, relative humidity and pressure were measured with a data logger (Votcraft DL-220THP, Votcraft®, Hirschau, Germany), biogas composition was evaluated in triplicate using a gas analyser (Biogas 5000; Geotech, Germany), and the biogas volume was monitored every day at 9:00AM using a gas flow meter (Gas Meter-JBD 2.5-SA; China) where the data is transferred to a computer and used for the normalization of the biogas and methane produced thereof.

Fermentation process

The fermentation process builds on two phases of experiments:

1. Substrate feeding with chopped stem
2. Substrate feeding with ground stem

Two digesters were filled to the level (until the liquid start to flow through the outlet) with 300 kg of fresh cow manure mixed with 300 L of water, as a seeding inoculum. Afterwards, one of the digesters were fed with substrates (fresh stem and water) while the other (control digester) is filled with equal volume but with water alone. Both digesters were fed on daily basis (Table 1). The daily feeding amount was calculated entailing the final concentration of the volatile solids of substrate in the digester to be 10 % and 7 % in the first and second phase of experiment, respectively.

Hydraulic retention time (HRT) is defined as the average time interval over which the substrate is kept inside the digester before displaced through the outlet. The organic load rate (OLR) is determined as organic matter (or volatile solids) feed-in per unit time and volume of the digester.

Experiment I: In the first phase of experiment the substrate was only the stem part of the WH fed to the digester while chopped to a length of 2–3 cm, without additional water. The fermentation was running for about 53 days.

Experiment II: In the second phase of experiment, the substrate underwent a size reduction (from 2–3 cm to < 1 cm) using mechanical grinder before feeding to the digester. The duration of fermentation process in this phase of experiment of the trial was 39 days. A portable biogas analyzer was used to measure the biogas produced from WH and inoculum using anaerobic digestion process [20].

Field Experimental design and treatments: The experiment was laid out in Randomized Complete Block Design (RCBD) with three replications as shown in Table 3. The experimental site was plowed two times and leveled before sowing. A plot size of 3 m x 2 m (6 m²) with 1.5 cm row spacing and a total of 3 rows were used. Experiment consisting of nine treatments as shown in the Table 2.

Results and discussion

General characterization of WH

Proximate analysis

The proximate analysis of WH biomass samples such as moisture content (MC), ash content (AC), volatile matter (VM) and fixed carbon content (FC) were determined (Table 4).

Moisture content of WH: As a consequence of the different moisture contents of different parts of the crop, the drying process is to separately organize to reach similar moisture contents of 10 %. The high moisture content of both plants samples indicate that they may

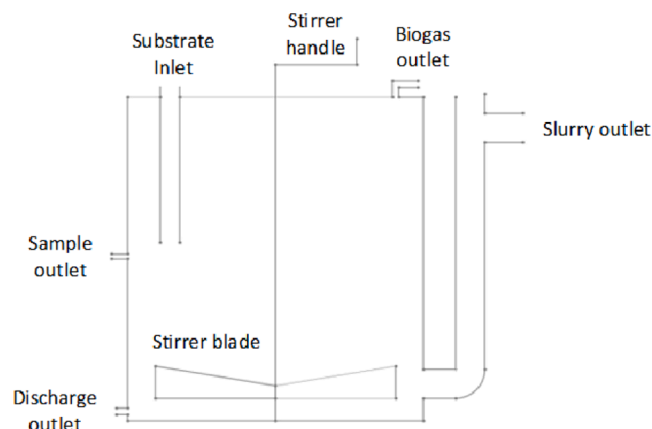


Fig. 2. Schematic picture of the pilot scale digester.

Table 1

Daily feeding of digester A (feedstock added) and digester B (without feedstock).

Phase of experiment	Digester A					Digester B
	Substrate* (g _{FM} /d)	Water (g/d)	Total volume added (g/d)	OLR (g _{VS} /L·d)	HRT (d)	
I	8646	0	8646	1.0	69	8646
	10,375	0	10,375	1.2	58	10,375
	12,104	0	12,104	1.4	50	12,104
II	3768	2232	6000	1.0	100	6000
	5652	3348	9000	1.5	67	9000
	7536	4464	12,000	2.0	50	12,000

* only the stem of water hyacinth.

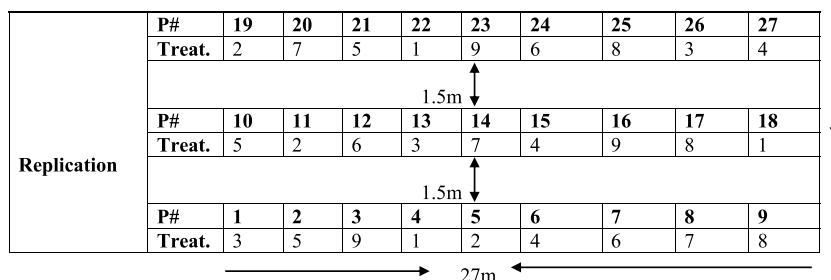
Table 2

Treatments with respective codes.

Treatments	Treatments Code	Level (Kg/ha)
Control	(T1)	0
Bio slurry ₁	(T2)	1000
Bio slurry ₂	(T3)	2500
Bio slurry ₃	(T4)	5000
Bio slurry ₄	(T5)	10,000
Water Hyacinth ₁	(T6)	2500
Water Hyacinth ₂	(T7)	5000
Water Hyacinth ₃	(T8)	10,000
Mixed Fertilizer	(T9)	1 + 1 (NPS+Urea)

Table 3

field layout of Teff crop.

**Table 4**

Proximate analysis of Water hyacinth (mean ±SD).

Sample	DM (%)	MC (%)	VS (%)	Ash (%)	FC (%)
Root	14.47±1.50	85.20±3.01	54.37±2.16	45.63±2.50	98.14±1.01
Stem	6.73±1.09	93.27±1.51	95.94±1.69	4.06±1.10	98.06±0.90
Leaf	14.47±1.53	88.70±1.25	95.95±1.76	4.05±1.68	98.14±1.03
FWP	8.87±0.98	91.13±1.93	91.85±1.19	8.15±1.42	98.08±0.86
DWP	91.90±1.92	8.10±1.22	63.35±2.26	36.65±1.59	98.91±0.62
Inoculum	9.40±0.57	90.51±2.68	60.16±3.49	39.84±2.04	98.09±0.14

DM: Dry Matter; MC: Moisture Content; VS: Volatile Solids; FC: Fixed Carbon.

be easily susceptible to spoilage if not well preserved [21].

Ash content: The root has seriously more ash content (inorganic residues after burning) than the other parts which indicates the higher presence of inorganic matter [22], which reduces the calorific value of the material.

Volatile solids: Solids, frequently organic, which volatilize at a temperature of 550 degrees Celsius. This is the actual organic matter which can be converted to gas [23]

Fixed carbon: The fixed carbon content is an indication of the amount of carbon present in the biomass precursor with a high

calorific value suitable for energy production [24], that is for biogas production.

Ultimate analysis of water hyacinth

To interpret the WH characteristics for biogas and bioslurry production several analyses have been done (Table 5). pH and EC values are consistently with what reported by [25]. Organic carbon, total nitrogen, hydrogen and total sulphur are quite comparable to other studies done by [26]. The biogas production from organic substrates mainly depends on the organic fraction degraded to CH₄ and CO₂. The carbon/nitrogen ratio (C/N) is an important factor influencing the biogas production. The C/N ratio is within the optimum range of 20–32 required for anaerobic digestion [27]. The water hyacinth biomass from Lake Tana has relatively high carbon to nitrogen ratio, a characteristic desired in substrates for biogas production (Table 5). Similar studies also observed a C/N of about 29.23 for the plant [28].

One reason that water hyacinths are so effective at removing excess nutrients is that they exhibit luxury consumption, particularly of phosphorus (Table 5). That is, they will absorb more of this nutrient than they can utilize for growth. Excess phosphorus is stored within the plant tissue. Phosphorus is scarce in most natural or agricultural soils. Therefore, the excess amount of phosphorus can be improved by the mass production and application of phosphorus-rich WH slurry.

Most heavy metal values are relatively high, comparable to that of many other crop species. However, no toxic levels of lead, silver and cadmium were detected in the plant tissue and even chromium concentration was very low (below detection limit). When detected at all, the amounts of these substances were no greater than those present in soybean and cotton seed meal [29]. The relatively high mineral content of the WH, comparable to that of many other crop species, suggests that this plant could make a good soil additive.

Biogas and bioslurry production and analysis at batch and semi-continuous experiments

An anaerobic batch experiment is a laboratory method used to determine the biodegradability and methane potential of organic materials under anaerobic conditions. The test involves adding a known quantity of the organic material to a reactor vessel (typically a bottle or jar) filled with an anaerobic inoculum (microbial community adapted to anaerobic conditions), and sealing the vessel to maintain anaerobic conditions. The reactor is then placed in a temperature-controlled environment (typically around 35–40 °C) to simulate the conditions of an anaerobic digester. Over a period of several weeks, the generated biogas (mainly methane and carbon dioxide) was collected and its biogas composition was evaluated in triplicate using a gas analyser (Biogas 5000; Geotech, Germany), and the biogas volume was monitored every day at 9:00 AM using a gas flow meter (Gas Meter-JBD 2.5-SA; China). The test can be used to determine the methane potential of the organic material, which represents the maximum amount of methane that can be produced through anaerobic digestion. The methane potential can be used to estimate the amount of biogas that could be produced from the organic material in a full-scale anaerobic digester and can be used to inform decisions on waste management and energy recovery.

The highest biogas and methane yield was obtained from the fresh stem, with values 597 and 326 L N/kg VS [litres (normalized) per kilogram of volatile solids], respectively (Table 6). In other explanation, 21 L N Methane could be recovered from a unit kilogram of fresh stem. Madiye reported that 1 kg of WH has yielded 12.1 L of biogas. The biogas/methane yield from same substrate, stem, but dried one was lower 427 L N/kg VS. This indicates that drying the stem of the WH and using it as a feed to a biogas digester will reduce the biogas yield of the substrate. However, there is a trade-off between transporting huge volume of WH with all its moisture or drying the WH after harvesting for longer transport and period of storage.

The biogas yield from the leaves and root of the WH was 311 and 225 L N/kg VS, respectively. A unit weight of fresh leaves and root (with moisture content of 88.7 % and 85.2 %) has a methane generating potential of 19 L and 10 L, respectively (Table 4, and Table 6). Common farm manures from cattle, pig and poultry generate methane in a range 130–300, 210–320 and 250–400 L N/kg VS, respectively. Agricultural products, like maize silage have a specific methane yield of 320–400 L N/kg VS, while grass generates 286–324 L N/kg VS [30]. The highest quality of the biogas (i.e., methane percentage) was obtained from the leaves, followed by the root and the stem. As Castro and Agblevor [31] reported an ultimate methane potential of water hyacinth obtained from the Ozama river (Dominican Republic) was 399.2 ± 32.2 L N/kg VS (all plant parts). The authors added that the estimated energy produced per ton of fresh biomass was 846.5 MJ.

About 50 per cent of the total methane yield from the fresh stem was recovered in less than a week from the commencement of the

Table 5
Mean and standard deviation of WH analysis (mean $\pm$ SD).

Parameter	Unit	Mean	Parameter	Unit	Mean
pH	–	6.1 $\pm$ 0.4	Fe	mg/kg	295.7 $\pm$ 1.60
C/N ratio	–	31.9 $\pm$ 1.0	Mn	mg/kg	27.6 $\pm$ 1.60
EC	ds/cm	3.2 $\pm$ 0.5	Cu	mg/kg	0.3 $\pm$ 0.03
OC	%	31.6 $\pm$ 2.5	Ni	mg/kg	0.2 $\pm$ 0.04
H	%	3.6 $\pm$ 0.2	Cd	mg/kg	0.1 $\pm$ 0.03
Nt	%	1.0 $\pm$ 0.4	Pb	mg/kg	0.3 $\pm$ 0.05
Pt	mg/kg	958.0 $\pm$ 3.6	Cr	mg/kg	ND
St	mg/kg	0.3 $\pm$ 0.1	Zn	mg/kg	0.8 $\pm$ 0.10
K	mg/kg	111.5 $\pm$ 0.9	As	mg/kg	1.1 $\pm$ 0.14
Ca	mg/kg	71.8 $\pm$ 2.6	Hg	mg/kg	0.1 $\pm$ 0.04
Mg	mg/kg	66.0 $\pm$ 1.8			

Table 6

Methane production potential of the WH substrates after 35 days of fermentations.

Sample type	Av. Methane content [% vol]	Av. Specific biogas yield [L_N/kg_{VS}]	Av. specific methane yield [$L_N CH_4/kg_{VS}$]	Av. specific methane yield [$L_N CH_4/kg_{TS}$]	Av. specific methane yield [$L_N CH_4/kg_{FM}$]
Fresh stem	55	597	326	265	21
Dry stem	52	427	224	190	170
Leaf	64	311	198	167	19
Root	58	225	131	65	10

anaerobic fermentation. The fermentation kinetics from the dried stem and fresh leaves showed nearly similar patterns. The root on the other side, showed fast fermentation, and thus three-quarters of the total methane was recovered in about one week time (Fig. 3).

The WH parts (root, stem, and leaf) contain varying amounts of easily digestible carbohydrates such as cellulose and hemicellulose, which are necessary substrates for methanogenic bacteria. Furthermore, the structure and surface area of each plant component affect the rate of decomposition and gas production. The stems of WH particularly rich in carbohydrates including cellulose and hemicellulose. These substrates are easily broken down by methanogenic Bacteria, resulting in significant methane production. The fiber, the volatile solid, the microbial activity and surface area of each plant part affects the methane yields of the substrate [32].

Biogas yield characterization at pilot scale

The highest performance of fermentation (excluding the biogas produced from the cow dung Inoculum) was achieved 28.4 L/kg_{FM} while feeding the digester with OLR 1.2 $kg_{VS}/m^3 \cdot d$ (Fig. 4). The average quality (per cent) of methane and carbon dioxide in the biogas was 59.2 ± 0.2 and 38.9 ± 0.2 , while the hydrogen sulphide showed a mere of 1.0 ± 0.8 ppm, respectively. Results of the present study well agrees with previous results obtained a mean value of 60.1 % methane (CH_4), as reported by Edimu et al. [33]. Thus, the highest methane yield can be calculated as 16.8 L/kg fresh stem. From the batch test, the specific methane yield from fresh stem was 20.6 L/kg_{FM} . This indicates that the performance of the pilot scale digesters slightly lower than the batch test. However, it is still worth using such a biomass resource obtained with no cost for the users.

The fermentation process achieved its highest performance, excluding biogas produced from cow dung inoculum, with a yield of 28.8 L of biogas per kilogram of fresh stem when the digester was fed with an organic loading rate (OLR) of 1.2 kg volatile solids (VS) per cubic meter per day (Fig. 4). The average methane and carbon dioxide content in the biogas was 59.2 ± 0.2 % and 38.9 ± 0.2 %, respectively, while hydrogen sulfide levels were minimal at 1.0 ± 0.8 ppm. These results align well with previous findings, which reported a mean methane content of 60.1 %, as documented by Edimu et al. [33]. Therefore, the highest methane yield can be estimated as 16.8 L per kilogram of fresh stem. In the batch test, the specific methane yield from fresh stem was 20.6 L per kilogram of fresh matter (FM), indicating slightly lower performance in the pilot scale digesters compared to the batch test. Nevertheless, utilizing such biomass resources obtained at no cost to users remains advantageous.

The biogas reactors was run in semi continuous mode for two phases with the average Water hyacinth (TS (18.10 %) MC (81.97 %) VS (87.97 %) and Ash (12.03 %)) and Phase two WH with the average (TS (8.64 %) MC (91.36 %) VS (80.32 %) Ash (19.68 %)) whereas the Control digester was with the average (TS (32.55 %) MC (67.45 %) VS (93.33 %) Ash (6.67 %))(210 kg cow dung + 210 L water)

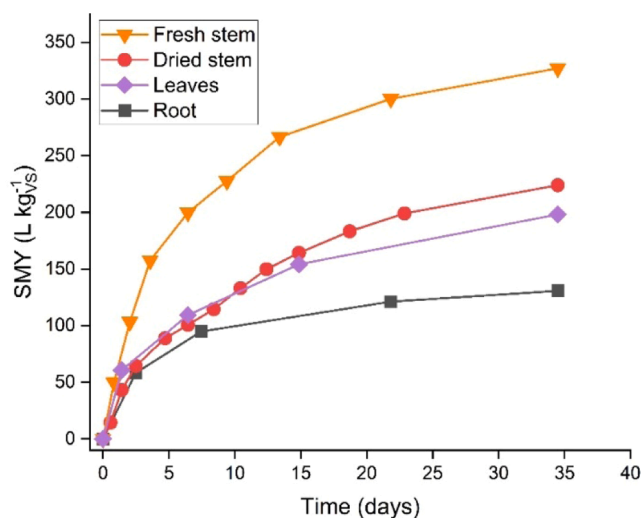


Fig. 3. Fermentation Kinetics of different components of the water hyacinth (fresh stem, dried stem, leaves and root of water hyacinth) specific methane yield [L/kg_{VS}] with time (days).

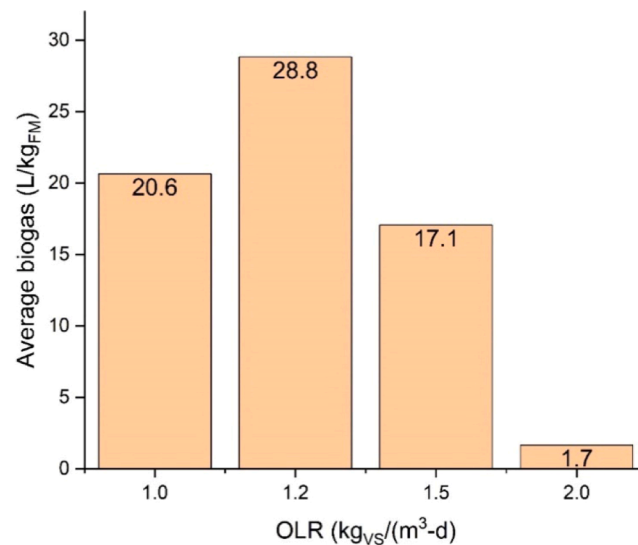


Fig. 4. Summarized specific biogas yield (0,5,10,15,20,25 and 30) L/kg_{FM} from phase I and Phase II of experiments with different organic loading rates (1.0,1.2,1.5 and 2.0) kg_{VS}/(m³-d).

was fed in the digester and kept for one week. this one week was actually the acclimatization period. After one week, feeding of the digester started at a very low OLR which was gradually increased at the mesophilic temperature (38 ± 2) and pH (7.3–7.54 & 7.3–7.43) for the two phase. The pH is in a range with Noxolo et al. [34] reported Methanogenic bacteria work effectively at the pH range of 6.5 to 8.

The biogas reactors were operated in semi-continuous mode for two phases. In the first phase, the average composition of Water Hyacinth (WH) was as follows: Total Solids (TS) 18.10 %, Moisture Content (MC) 81.97 %, Volatile Solids (VS) 87.97 %, and Ash 12.03 %. For the second phase, the average composition of WH was slightly different, with TS 8.64 %, MC 91.36 %, VS 80.32 %, and Ash 19.68 %. In contrast, the Control digester had an average composition of TS 32.55 %, MC 67.45 %, VS 93.33 %, and Ash 6.67 %. Each digester was fed with a mixture of 210 kg cow dung and 210 L water, which was then left in the digester for one week to acclimatize.

During this acclimatization period, the feeding of the digester began at a very low Organic Loading Rate (OLR), which was gradually increased while maintaining a mesophilic temperature range of 38 ± 2 °C and pH levels ranging from 7.3 to 7.54 in the first phase, and 7.3 to 7.43 in the second phase.

These pH values are consistent with those reported by Noxolo et al. [35], who found that methanogenic bacteria work effectively within the pH range of 6.5 to 8. A linear increase in biogas production was observed at the lower OLR 1 (kgVS m³ d⁻¹). However when the OLR gradually increases the reduction in biogas production was observed, suggesting that the activity of methanogenic bacteria was not impaired up to OLR values of 1 kgVS m³ d⁻¹. Similar to phase one there was reduction in biogas in phase two when the OLR increases above 1.2 kgVS m³ d⁻¹ as shown in Fig. 4. This result is similar to those reported by Jayaweera [36], Barua & Kalamdhad [10].

A linear increase in biogas production was observed at lower Organic Loading Rates (OLR), specifically at 1 (kgVS m³ d⁻¹). However, as the OLR gradually increased, a reduction in biogas production was observed. This suggests that the activity of methanogenic bacteria was not impaired up to OLR values of 1 kg VS m³ d⁻¹. Similarly, in phase two, there was a reduction in biogas production when the OLR increased above 1.2 kgVS m³ d⁻¹, as depicted in Fig. 4.

These findings are consistent with those reported by Jayaweera [36], Barua & Kalamdhad [10].

Anaerobic digestion of cow dung (blank) has a low biogas production due to the long hydraulic retention time (HRT), co-digestion with water hyacinth is effective at improving the yield biogas production. Lowering the HRT can therefore increase the methanogenic efficiency during production. The highest methane production was achieved with an HRT of 6–10 days. The maximum methane yields (834 l/d) were attained at HRT ≤ 15 days and increasing the HRT to 30 days resulted in low methane yields and complete process failure at ≥ 30 days HRT, due to the blockage of the digester and microorganisms washed out. The results by Shaojie [37] also indicated that there is a correlation between microbial community and HRT could be utilized as a bio-indicator for better process performance and to foresee process failure.

Anaerobic digestion of cow dung alone typically results in low biogas production due to the long hydraulic retention time (HRT). However, co-digestion with water hyacinth has proven effective in improving biogas production yields. Lowering the HRT can enhance methanogenic efficiency during production. The highest methane production was achieved with an HRT of 6–10 days. Maximum methane yields (834 L/d) were attained at HRT ≤ 15 days. Increasing the HRT to 30 days resulted in lower methane yields and complete process failure at ≥ 30 days HRT due to digester blockage and washout of microorganisms. Results from Shaojie et al. [37] suggest a correlation between microbial community and HRT, which could serve as a bio-indicator for better process performance and predicting process failure.

Biogas production from water hyacinth involves anaerobic digestion (a process of microorganisms breakdown organic matter in absence of oxygen) results in methane and carbon dioxide. With its high organic matter concentration, WH is an ideal feedstock for the AD process. This process includes hydrolysis (polymers are broken down into simple sugars), acidogenesis (simple sugars are converted into volatile fatty acids VFAs), acetogenesis (VFAs are converted into acetic acid), and lastly methanogenesis (methanogenic archaea use acetic acid to create CH_4 and CO_2). The first component that affects the production of biogas is the feedstock composition (cellulose, hemicellulose, and lignin), followed by the pretreatment methods (milling) and biological pretreatment (using inoculum), digester parameters (temperature, pH, HRT), and stirring. As the microbes break down the WH, vital elements including nitrogen, phosphorus, and potassium are released into the digestate, which are crucial for plant development. Because of this, the resultant bioslurry is a nutrient-rich organic fertilizer that may be used to increase soil fertility and crop yields [38].

Analyses of bio slurry derived from water hyacinth

The results of the physicochemical analysis are summarized in Table 7. It is worth noting that the metal values obtained in this study are significantly lower compared to the findings of other studies [39].

Slurry of biogas (bio slurry), which is a by-product of biogas technology can be utilized as a source of organic soil material like manure. Bioslurry is useful for the fertilization of cereals and maize, fruit trees and alley crops as well as pasture and forage land. Bioslurry from biogas has an advantage when compared to manure or compost, due to the fact that the nutrients can be directly exploited by plants [40].

The main causes for nutrient imbalance of the bioslurry are inadequate composition of the feedstock, the storage conditions create the loss of nutrients through volatilization [41] and decomposition. The other is inappropriate application methods for instance applying the bioslurry on the soil surface lead to nutrient loss through leaching [42] there are various factors for electrical conductivity and macrobial activity stress in bioslurry including extreme temperature, changes in pH, presence of pollutants like heavy metals those factors can alter the microbial activity. The electrical conductivity stress of the bioslurry mainly due to the increased concentration of dissolved salts and nutrients, which can create the osmotic stress and reduce water availability to plants [43].

Crop application of bio slurry (Agronomic trials)

The results of the field trial evaluation are summarized in Table 8. At a confidence level of $p < 0.05$, treatment two (BS_1) displayed the highest plant height (69.93 ± 1.00) and the longest panicle length ($34.53 \text{ cm} \pm 4.06$). Furthermore, treatment five (BS_4) exhibited the highest tiller number, followed by treatment nine (MF), while treatment two (BS_2) showed the third-highest effective tiller count at $p < 0.05$, with a value of (7.6 ± 0.80).

At a confidence level of $p < 0.05$, treatment two (BS_1) exhibited the highest plant height was recorded from 1000 (Kg/ha) (69.93 ± 1.00) cm and absolute control treatments, while the lowest plant height of 48.00 ± 2.45 cm was recorded from 10,000 (Kg/ha). The result of this study became agreed with the highest plant height obtained at the higher blended fertilizer levels and increases with enhancing the amount of blended fertilizer rate which might be because of the vital role of N applied for elongation and vegetative growth [44].

Analysis of variance at $p < 0.05$ revealed that panicle length of teff was not significantly influenced by treatment rates. The highest panicle length from treatment BS_1 of ($34.53 \text{ cm} \pm 4.06$) cm, while the lowest panicle length of 28.26 ± 2.30 cm was measured from the absolute WH3 treatment. The result of this study is in line of agreement with increasing application of bioslurry significantly increased panicle length, but further increasing did not consistently increase panicle length [45]. Teff plants with higher panicle length were exhibited by applying (1000 Kg/ha) bioslurry due to high nitrogen usually favors vegetative growth of teff, which reveals in relatively greater panicle length [46]. Mixed fertilizer types and rates (50 NPS + 50 Urea) had no significant effect on panicle length of teff and this result was in the line of agreement with panicle length of teff did not influence by mixed fertilizer rather than by N application [47].

Analysis of variance revealed that the total tiller number per plant of teff was significantly influenced by fertilizer 13.80 ± 2.20 from (50 NPS + 50 Urea). The lowest total tiller number per plant at $p < 0.05$, with a value of 6.40 ± 1.24 in WH3 was recorded from unfertilized treatment. Similarly, Tewolde et al. [44] reported that there was no significant difference except with the control plot on both

Table 7
Physicochemical analysis of bioslurry form of water hyacinth.

Parameters	unit	This study	Parameters	unit	This study	[39]
pH	–	7.32 ± 0.9	Fe	mg/kg	271 ± 3.3	6154
C/N ratio	–	11.65 ± 0.9	Mn	mg/kg	6.81 ± 0.1	471
EC	ds/cm	10.42 ± 1.5	Cu	mg/kg	0.2 ± 0.0	400
OC	%	12.52 ± 2.5	Ni	mg/kg	0.5 ± 0.2	62
TN		1.08 ± 0.2	Cd	mg/kg	0.05 ± 0.0	3
TP	mg/kg	10.8 ± 0.03	Pb	mg/kg	ND	150
TS	mg/kg	87.8 ± 0.3	Cr	mg/kg	ND	210
K	mg/kg	143.5 ± 3.1	Zn	mg/kg	1.0 ± 0.01	700
Ca	mg/kg	108.9 ± 0.6				
Mg	mg/kg	69.2 ± 0.2				

Table 8
Effect of bioslurry and inorganic fertilizers on Teff productivity (mean±SD).

Treatments	Level (Kg/ha)	Plant Height (cm)	Panicle length (cm)	Number of Nodes	Lower internode length	total tiller	Effective Tiller	Grain yield (kg/ha)
Control	0	63.33 ±2.08	30.93±2.13	3.20±0.00	7.50±2.75	13.26 ±3.64	6.20±1.40	88.88±3.51
BS1	1000	69.93 ±1.00	34.53±4.06	3.66±0.23	9.86±1.28	11.06 ±0.98	7.26±0.11	158.33 ±5.00
BS2	2500	64.33 ±0.75	34.06±1.10	3.06±0.11	9.53±2.65	12.00 ±2.30	7.60 ± 0.80	169.43 ±3.51
BS3	5000	60.20 ±1.24	29.60±1.63	2.73±0.50	6.96±1.22	10.86 ±2.99	7.26±0.98	144.41 ±3.51
BS4	10,000	48.00 ±2.45	33.00±3.51	3.06±0.30	9.56±2.23	12.06 ±2.48	6.40±1.74	116.67 ±5.00
WH1	2500	67.20 ±2.00	33.26±4.27	3.40±0.20	9.33±0.80	13.00 ±1.90	6.13±0.23	100±5.00
WH2	5000	56.46 ±3.12	29.40±3.27	2.60±0.40	6.83±1.43	12.93 ±2.73	5.93±0.64	100±1.00
WH3	10,000	53.53 ±3.44	28.26±2.30	2.93±0.46	6.86±1.34	6.400 ±1.24	6.40±1.24	61.67±1.00
MF	(50 NPS +50Urea)	62.60 ±1.63	31.53±0.75	3.20±0.34	7.40±1.90	13.80 ±2.20	6.80±1.00	159.43 ±5.03
p-value		<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001

the total number of tillers and productive/effective tillers. The current result agrees with total and fertile tiller increased consistently and significantly in response to increasing the rate of fertilizer [48]; increasing organic and inorganic fertilizers levels resulted in linear and consistent increment of tiller number while, the main effect of mixed fertilizer did not affect tiller number of teff [47] at $p < 0.05$, Analysis of variance revealed that grain yield of teff was significantly influenced by blended fertilizer types and rates. The highest grain yield of BS₂ (169.43 kg/ha±3.5) was recorded from the application of 2500 Kg/ha, while the lowest grain yield of (61.67 kg/ha±1.00) was recorded from unfertilized treatment as shown in Table 8, the increased grain yield might be due to effect of balanced nutrients on improving crops agronomic performance thereby enhancing nutrient use efficiency [49] and The grain yield of Teff significantly increased with the application of bioslurry at ($p \leq 0.05$) compared to mixed fertilizer and the control (Table 8). As Fig. 5 shows, the grain yield increment from plot treated with bioslurry and mixed fertilizer might be due to the contribution of balanced nutrient (macro and micronutrient) present in fertilizers which increased yield attributes through more uptakes of all the nutrients and increased translocation of photosynthetic materials from source to sink [47]

Previous studies, such as Hossain et al., reported a positive effect of bioslurry on improving yield, with the highest rate of 78 qun/ha proving to be optimal for enhancing vegetative growth. Alamgir & Kibria also reported a significant improvement in yield with bioslurry application. The results suggest that the application of biogas slurry on *Eragrostis Teff* positively influences its yield, with increased yield and yield parameters observed through bioslurry and mixed fertilizer applications. treatments BS₂ performed better than others BS₄, despite higher application rates it is due to higher application rate of bio slurry leads to nutrient imbalance and reduce grain yields, especially when the application rate exceeds the plant needs or soil capacity to absorb and utilize the nutrients. The findings recommend the use of 2500–5000 kg/ha of bioslurry for significantly improved Teff grain yield. Further research exploring different rates under various soil and agro-ecological zone conditions would be beneficial in determining specific rates suitable for each production area.

Fig. 5.

Conclusion

The comprehensive exploration of WH for biogas and bioslurry production, along with Teff crop yield optimization, has provided valuable insights into the potential of this renewable resource. The batch and semi-continuous experiments allowed for a detailed understanding of the anaerobic digestion process and the biogas yield of different WH components. The results from batch experiments highlighted the fresh stem of water hyacinth as a key contributor to both biogas and bioslurry production. This information is pivotal for optimizing resource utilization and enhancing the overall sustainability of the process. Proximate and ultimate analyses of biomass components further contributed to our knowledge of the chemical composition, facilitating better process refinement. The semi-continuous experiments, carried out in stainless steel digesters, demonstrated the feasibility of integrating WH into a continuous biogas and bioslurry production system. Monitoring key parameters provided crucial data for process optimization and stability. The highest biogas and methane yields obtained from the fresh stem underscore the importance of targeted utilization for specific components. Further Research should be continued to explore additional factors influencing the efficiency of water hyacinth-based biogas and bioslurry production. Investigate the potential of different pre-treatment methods to enhance the breakdown of WH biomass. In conclusion, the findings from this study lay a foundation for the sustainable utilization of water hyacinth, offering a promising avenue for biogas and bioslurry production. By addressing the outlined recommendations, future research can contribute to the advancement and widespread adoption of this eco-friendly and resource-efficient approach.

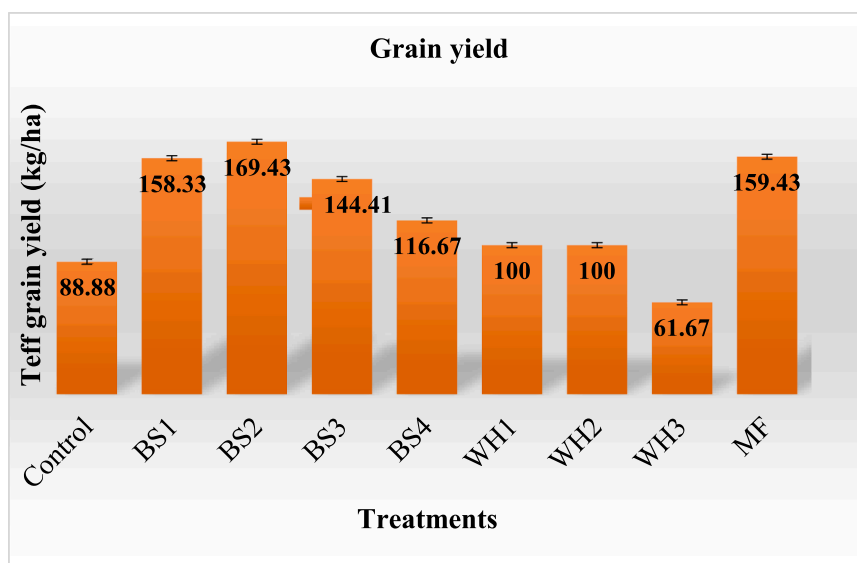


Fig. 5. The effect of WH bioslurry and chemical fertilizer on eragrotis teff grain yield soil nutrients across different bioslurry treatments. The treatments are control, Bioslurry one (BS1), bioslurry two (BS2), bioslurry three (BS3), bioslurry four (BS4), water hyacinth one (WH1) water hyacinth two (WH2), water hyacinth (WH3) and mixed fertilizer (MF).

CRediT authorship contribution statement

Tewodros Mulu: Conceptualization, Data curation, Software, Visualization, Writing – original draft. **Bilhate Chala:** Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Software, Supervision. **Yezbie Kassa:** Formal analysis, Software. **Tayachew Nega:** Formal analysis, Writing – review & editing. **Maregn Adugna:** Writing – review & editing. **Mohamed Gedefaw:** Writing – review & editing. **Teferi Alem:** Writing – review & editing. **Kiflie Tesfamariam:** Visualization, Writing – original draft. **Taddesse Adgo:** Writing – review & editing. **Beatriz Waldman:** Funding acquisition, Project administration, Resources, Validation, Writing – review & editing. **Bernhard Freyer:** Funding acquisition, Investigation, Methodology, Project administration, Resources, Supervision, Validation, Writing – review & editing. **Dessie Tibebe:** Conceptualization, Funding acquisition, Investigation, Methodology, Project administration, Supervision, Writing – review & editing.

Declaration of interests

The authors declare that we have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Declaration

Ethical approval and guidelines: “All methods are carried out according to the institution’s guidelines and regulations”.

Consent for publication: “Not Applicable”.

Availability of data and materials: All data generated and analyzed are included within this research article.

Funding: This project work was funded by NABU.

Acknowledgment: The authors gratefully acknowledge NABU and University of Gondar for supporting the work.

References

- [1] A. Fagerström, J.D. Murphy, IEA Bioenergy Task 37., & IEA Bioenergy Programme, *The role of anaerobic digestion and biogas in the circular economy*, IEA Bioenergy 8 (2) (2020) 1–24.
- [2] V. Chaturvedi, P. Verma, An overview of key pretreatment processes employed for bioconversion of lignocellulosic biomass into biofuels and value added products, 3 Biotech 3 (5) (2013) 415–431, <https://doi.org/10.1007/s13205-013-0167-8>.
- [3] Kumar, G., Mehmood, T., & Cheng, L. (2020). *Water hyacinth as a biomass : a review*. 277. doi:10.1016/j.jclepro.2020.122214.
- [4] V. Guna, M. Ilangoan, M.G. Anantha Prasad, N. Reddy, Water hyacinth: a unique source for sustainable materials and products, ACS Sustain. Chem. Eng. 5 (6) (2017) 4478–4490, <https://doi.org/10.1021/acssuschemeng.7b00051>.
- [5] D. Amare, W. Mekuria, B. Belay, Willingness and participation of local communities to manage communal grazing lands in the Lake Tana Biosphere, Ethiopia, Soc. Nat. Resour. 30 (6) (2017) 674–689, <https://doi.org/10.1080/08941920.2016.1264649>.
- [6] Luba. (2023). *The role of biogas production in circular economy approach from the perspective of locality*.
- [7] Dessie, K. Jembere, A. Kidie, M. Adugna, T. Alem, G. Teshome, Compost preparation, chemical analyses and users' perception in the utilization of water, BMC Chem. (2022) 1–10, <https://doi.org/10.1186/s13065-022-00851-9>.

- [8] Paul Njogu, Robert Kinyua, Y.N. Purity Muthoni, Environmental implication of over population and rural-urban migration on development in Nigeria, *Energy Power Eng.* 07 (05) (2013) 209–216, <https://doi.org/10.4236/epe.2015.75021>.
- [9] Fadaio, Biogas production from water hyacinth blends, in: *10th International Conference on Heat Transfer, Fluid Mechanics and Thermodynamics, Orlando Florida, July, 2014*, pp. 792–799. https://repository.up.ac.za/bitstream/handle/2263/44610/Fagbenle-Biogas_2014.pdf.
- [10] M.W. Jayaweera, J.A.T. Dilhani, R.K.A. Kularatne, S.L.J. Wijeyekoon, Biogas production from water hyacinth (*Eichhornia crassipes* (Mart.) Solms) grown under different nitrogen concentrations, *J. Environ. Sci. Health - Toxic/Hazard. Subst. Environ. Eng.* 42 (7) (2007) 925–932, <https://doi.org/10.1080/10934520701369842>.
- [11] M.P. Maria, N.H. Torres, V.R.S. Nascimento, T.S.A. Chagas, G.D. Saratale, S.I. Mulla, R.N. Bharagava, E.B. Cavalcanti, L.F. Romanholo Ferreira, Current advances in the brewery wastewater treatment from anaerobic digestion for biogas production: a systematic review, *Environ. Adv.* 13 (March) (2023), <https://doi.org/10.1016/j.envadv.2023.100394>.
- [12] T. Kebede, Y.G. Keneni, A.F. Senbeta, G. Sime, Effect of bioslurry and chemical fertilizer on the agronomic performances of maize, *Heliyon* 9 (1) (2023) e13000, <https://doi.org/10.1016/j.heliyon.2023.e13000>.
- [13] C. Tamiru, The origin, variation, immunity and breeding of cultivated plants selected writings, *Int. J. Hortic. Agric. Food Sci.* 12–13 (3) (2021) 14–24, <https://doi.org/10.22161/ijhaf.5.3.3>.
- [14] A. Arefaine, D. Adhanom, N. Tekeste, Response of teff (*Eragrostis tef* (Zucc.) Trotter) to seeding rate and methods of sowing on yield and yield attributes in a subhumid environment, Northern Ethiopia, *Int. J. Agron.* 2020 (2020), <https://doi.org/10.1155/2020/1516790>.
- [15] FAO. (2021). *Food and agriculture making agrifood systems more resilient to shocks*.
- [16] FAO. (2021). *Standard operating procedure for soil available phosphorus*.
- [17] W. Willson & Aust. J. Agric. Res., *J. Agric. Res.* 10 (1959) 340–352.
- [18] AOAC. (1990). *AOAC: official methods of analysis (Volume 1)*. 1(Volume 1).
- [19] C. Wan, Y. Li, Bioresource Technology effectiveness of microbial pretreatment by ceriporiopsis subvermispora on different biomass feedstocks, *Bioresour. Technol.*, *Technol* 102 (16) (2011) 7507–7512, <https://doi.org/10.1016/j.biortech.2011.05.026>.
- [20] Geotec. (2020). *BIOGAS 5000 gas analyzer operating manual*. 800, 1–61.
- [21] Ketema Seyfu, *Germplasm evaluation and breeding work on teff (Eragrostis tef) in Ethiopia*, *J. Appl. Sci. Env., Manage* (2009).
- [22] Sukarni, S., Zakaria, Y., Sumarli, S., & Wulandari, R. (2019). *Physical and chemical properties of water hyacinth (Eichhornia crassipes) as a sustainable biofuel feedstock Physical and chemical properties of water hyacinth (Eichhornia crassipes) as a sustainable biofuel feedstock*. doi:10.1088/1757-899X/515/1/012070.
- [23] A.O. Jimoh, M.M. Namadi, K. Ado, Proximate and ultimate analysis of *Eichhornia natans* (Water Hyacinth), *Pistia stratiotes* (Water Lettuce) and *Nymphaea lotus* (Water Lily) in the production of biofuel, *Pelagia Res. Libr.* 7 (4) (2016) 243–249.
- [24] Wauton*, William-Ebi, Characterization of water hyacinth (*Eichhornia Crassipes*) for the production of thermochemical fuels, *J. Multidiscip. Eng. Sci. Stud.* (JMSESS) 5 (7) (2019) 2458–2925. www.jmss.org.
- [25] Rai, U., Bharath, Y., Shetty, V., Patil, J.H., & Raj, M.A. (2020). *Kinetics of Anaerobic digestion of water hyacinth using primary sludge As inoculum*. December. doi:10.1109/GTEC.2011.6167658.
- [26] S. Soeprijanto, I.D.A.A. Wirmadewanthi, M.S. Muntini, A. Anzip, The utilization of water hyacinth for biogas production in a plug flow anaerobic digester, *Int. J. Renew. Energy Dev.* 10 (1) (2020) 27–35, <https://doi.org/10.14710/ijred.2021.21843>.
- [27] Ganguly, A., Das, S., Bhattacharya, A., Dey, A., & Chatterjee, P.K. (2020). *Enzymatic hydrolysis of water hyacinth biomass for the production of ethanol : optimization of driving parameters Enzymatic hydrolysis of water hyacinth biomass for the production of ethanol : optimization of driving parameters*. August 2015.
- [28] A.K. Mathew, I. Bhui, S.N. Banerjee, R. Goswami, A.K. Chakraborty, A. Shome, S. Balachandran, S. Chaudhury, Biogas production from locally available aquatic weeds of Santiniketan through anaerobic digestion, *Clean Technol. Environ. Policy* 17 (6) (2020) 1681–1688, <https://doi.org/10.1007/s10098-014-0877-6>.
- [29] B.C. Wolverton, R.C. McDonald, Nutritional composition of water hyacinths grown on domestic sewage, *Econ. Bot.*, *Bot* 32 (4) (2020) 363–370, <https://doi.org/10.1007/BF02907930>.
- [30] B. Chala, H. Oechsner, S. Latif, J. Müller, Biogas potential of coffee processing waste in Ethiopia, *Sustainability* 10 (8) (2018) 1–14, <https://doi.org/10.3390/su10082678>.
- [31] Castro, Y.A., & Agblevor, F.A. (2020). *Biomethanation of invasive water hyacinth from eutrophic waters as a post weed management practice in the Dominican Republic : a developing country*.
- [32] Q. Struik, E.S. Oliveira, A.J. Veraart, S. Kosten, Methane emissions through water hyacinth are controlled by plant traits and environmental conditions, *Aquat. Bot.*, *Bot* 183 (September) (2022) 103574, <https://doi.org/10.1016/j.aquabot.2022.103574>.
- [33] M.G.G. Edimu, D.A.M. Römer, D.J. Roberts, Effect of genotype and feeding regime on enteric methane, non-milk nitrogen and performance of dairy cows during the winter feeding period, *Livest. Sci.* 122 (2–3) (2018) 323–332, <https://doi.org/10.1016/j.livsci.2008.09.020>.
- [34] S.T. Noxolo, M. Edison, H.B. Tesfagiorgis, Effect of temperature and pH on the Anaerobic digestion of grass silage, in: *Proceedings of the 6th International Conference on Green Technology, Renewable Energy and Environmental Engineering*, Cape Town, South Africa, November, 27–28, 2014.
- [35] Pelton, A.R., & Integrity, S. (2014). *Effect of temperature and pH on the corrosion resistance of*. January 2005, 6–9.
- [36] V.B. Barua, A.S. Kalamdhad, Biogas production from water hyacinth in a novel anaerobic digester: a continuous study, *Process Saf. Environ. Prot.* 127 (2019) 82–89, <https://doi.org/10.1016/j.psep.2019.05.007>.
- [37] Shaojie, X. Hong, H. Yang, X. Yu, S. Fang, Y. Bai, J. Liu, Y. Gao, L. Yan, W. Wang, Y. Wang, Effect of hydraulic retention time on anaerobic co-digestion of cattle manure and food waste, *Renew. Energy* 150 (2020) 213–220, <https://doi.org/10.1016/j.renene.2019.12.091>.
- [38] E.O. Auma, PhD Thesis, University of Nairobi, 2007, pp. 1–134.
- [39] Environment, C.C. of M. of the. (2005). *CCME Guidelines for Compost quality*. 26.
- [40] S.L. Arnold, J.W. Doran, J. Schepers, B. Wienhold, D. Ginting, B. Amos, S. Gomes, Portable probes to measure electrical conductivity and soil quality in the field, *Commun. Soil Sci. Plant Anal. Soil Sci. Plant Anal.* 36 (15–16) (2005) 2271–2287, <https://doi.org/10.1080/00103620500196689>.
- [41] Bogdanski. (2013). *A review of scientific literature on the co-product of biogas production*.
- [42] Haile, A., & Ayalew, T. (2018). *Comparative study on the effect of bio-slurry and inorganic N-fertilizer on growth and yield of kale (Brassica oleracea L.)*. 12(April), 81–87. doi:10.5897/AJPS2018.1639.
- [43] Dey, A., & Kushwah, A. (2023). *Temporal characterization of biogas slurry : a pre-requisite for sustainable nutrification in crop production*. September 2021–2022. doi:10.3389/fsufs.2023.1234472.
- [44] B. Tewelde, G. Gebreyohannes, K. Abhra, Validation of blended NPSB fertilizer rates on yield, yield components of Teff [*Eragrostis tef* (Zuccagni) Trotter] at vertisols of Hatesbo, Central Tigray, Ethiopia, *J. Soil Sci. Environ. Manag.* 11 (2) (2020) 75–86, <https://doi.org/10.5897/jsem2019.0795>.
- [45] O. Giday, H. Gibrekidan, T. Berhe, Response of teff (*Eragrostis tef*) to different rates of slow release and conventional urea fertilizers in vertisols of southern Tigray, Ethiopia 1 (5) (2014) 190–197, <https://doi.org/10.15406/apar.2014.01.00030>.
- [46] H. Gebretsadik, M. Haile, C.F. Yamoah, *Tillage frequency, soil compaction and N-fertilizer rate effects on yield of teff (Eragrostis Tef (Zucc) trotter) in Central zone of Tigray, Northern* 1 (1) (2009) 82–94.
- [47] Mesfin, T., Haile, W., & Kassa, S. (2019). *Effects of nitrogen and blended fertilizers on yield and yield components of teff [(Eragrostis tef (Zucc.) Trotter)] in central highlands of Ethiopia*. 6(8), 15–64.
- [48] W. Tesfahun, Teff yield response to NPS fertilizer and methods of sowing in East Shewa, Ethiopia, *J. Agric. Sci. - Sri Lanka* 13 (2) (2018) 162–173, <https://doi.org/10.4038/jas.v13i2.8340>.
- [49] F. Asefa, A. Debelu, M. Mohammed, Evaluation of teff [*Eragrostis tef* (Zuccagni) Trotter] responses to different rates of NPK along with Zn and B in Didessa district, Southwest. Ethiop. 32 (11) (2014) 2245–2249, <https://doi.org/10.5829/idosi.wasj.2014.32.11.14562>.